

Case Report

A 15-Year-Old Boy with Primary Maxillary Bone Anaplastic Lymphoma Kinase-Positive Anaplastic Large Cell Lymphoma Relapsed with Rib Metastasis after Spontaneous Remission of a Maxillary Bone Lesion: A Case Report and Literature Review

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Keywords

Anaplastic large cell lymphoma · Spontaneous remission · Primary bone lymphoma

Abstract

Anaplastic large cell lymphoma (ALCL) is a rare form of non-Hodgkin's lymphoma (NHL) in children, accounting for 10–15% of all NHL cases. ALCL is currently classified as follows: systemic anaplastic lymphoma kinase (ALK)-positive, systemic ALK-negative, primary cutaneous, and breast implant-associated ALCL. In children, systemic ALK-positive ALCL is the most common, and patients often present with extranodal involvement. We report a rare case of systemic ALK-positive ALCL with primary bone involvement in a 15-year-old male patient. Primary bone lymphoma is most commonly observed in diffuse large B-cell lymphoma and is extremely rare in systemic ALCL. Therefore, the clinical features and prognosis of primary bone ALCL remain unclear. Our patient had spontaneous remission of primary maxillary bone ALCL after gingival scraping but relapsed 12 months later with rib metastasis. Spontaneous remission of ALCL has been reported frequently in primary cutaneous ALCL and rarely in systemic ALCL. Our case demonstrates for the first time that systemic ALCL can also present as solitary bone involvement that can spontaneously remit. Because systemic ALCL is aggressive and has a risk of relapse, as in

our case, it is important to consider ALCL in the differential diagnosis of primary bone lesions and to make a precise pathological diagnosis.

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Introduction

Anaplastic large cell lymphoma (ALCL) is a rare form of non-Hodgkin's lymphoma (NHL) in children, accounting for 10–15% of all NHL cases in the pediatric population [1]. According to the World Health Organization lymphoma classification, there are four types of ALCL: systemic anaplastic lymphoma kinase (ALK)-positive ALCL, systemic ALK-negative ALCL, primary cutaneous ALCL, and breast implant-associated ALCL. In children, 90% of ALCL cases are estimated to be systemic ALK positive [1], and patients frequently present with extranodal involvement, noting that they are often diagnosed at an advanced stage. Systemic ALCL with primary bone lesions presenting as extranodal involvement is very rare [2]. This disease is an aggressive lymphoma, usually treated by multidrug chemotherapy, with only a few reports of spontaneous remission [3–7]. There are no reports of spontaneous remission of primary bone ALCL in systemic ALCL cases. Here, we report a case in which a maxillary lesion showed spontaneous remission after gingival scraping; however, the patient relapsed with rib metastasis 12 months later, leading to a diagnosis of primary bone ALK-positive ALCL. The CARE Checklist has been completed by the authors for this case report, attached as online supplementary material (for all online suppl. material, see www.karger.com/doi/10.1159/000530459).

Case Presentation

A 15-year-old male patient presented to a community hospital with left-sided chest pain lasting longer than 4 weeks. Chest computed tomography (CT) and magnetic resonance imaging (MRI) revealed bone destruction and a dark mass in the left fifth rib. The patient was referred to our hospital for evaluation of the mass. The patient had no family history of cancer at a young age. Approximately 12 months prior to admission, he was diagnosed with buccal cellulitis, maxillary sinusitis, and periapical periodontitis at the root of the left maxillary first molar and second premolar. No apparent originating organism was reported in the bacteriological examination. The patient was subsequently treated with antibiotics for 14 days. However, gingival swelling and fever did not improve. A head CT scan showed a left maxillary lesion with bone destruction (Fig. 1a, left). Intraoperative findings revealed that the buccal bone of the maxillary first and second premolars was deficient, the gingival granulation tissue of the defect was scraped, and no bone removal or drilling was performed. Inflammatory cells were identified in the pathological specimens of the scraped granulation tissue. After the gingival granulation tissue was scraped, the patient recovered from fever, and a second CT scan 8 months after scraping revealed normal findings (Fig. 1a right).

On admission, the patient had no fever or lesions in the oral cavity. The left region of his chest was slightly warm but not swollen. No enlarged lymph nodes, hepatosplenomegaly, or rashes were observed. The patient's serum C-reactive protein was 3.28 mg/dL (normal level, <0.20 mg/dL) and soluble interleukin-2 receptor level was 2,300 U/mL (normal level, 121–613 U/mL). A bone marrow biopsy revealed normocellular marrow without lymphoma cell infiltration. Contrast-enhanced MRI was performed on admission and confirmed the

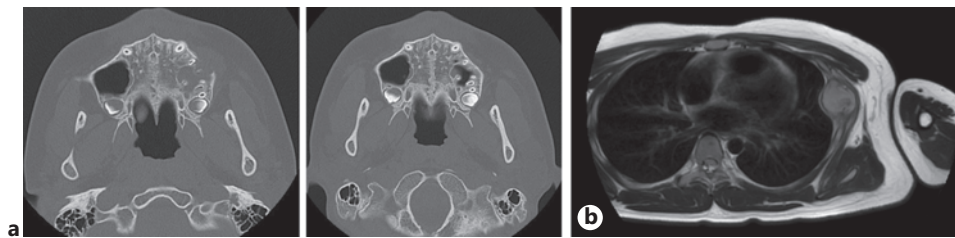


Fig. 1. Images at diagnosis of primary and recurrent diseases. **a** Computed tomography (CT) imaging before scraping surgery at another hospital 12 months before admission showing a maxillary sinus lesion with bone destruction (left). CT 8 months after scraping surgery, which turned normal (right). **b** Contrast-enhanced MRI on admission to our hospital. The lesion was up to 30 × 30 mm in diameter on the left fifth rib. T2-weighted images were hypointense.

presence of a lesion up to 30 × 30 mm in diameter on the left fifth rib. The tumor was hypointense on both T1- and T2-weighted images (Fig. 1b). Fluorodeoxyglucose-positron emission tomography combined with CT showed high fluorodeoxyglucose uptake (maximum standardized value of 26.9) only in the left fifth rib and no uptake in the left maxillary lesion.

We reviewed the pathological specimens of the left fifth rib lesion. In the hematoxylin and eosin-stained sections, we detected large atypical cells with densely stained, swollen nuclei, abundant eosinophilic cytoplasm, and occasional mitotic figures (Fig. 2a). These tumor cells were immunoreactive for CD30 (Fig. 2b), epithelial membrane antigen, and granzyme B. Their nuclei and cytoplasm were positive for ALK1 immunostaining (Fig. 2b) and negative for CD20 and CD79a. Reexamination of the pathological specimen of the left maxillary bone lesion revealed a sheet-like proliferation pattern of large cells with round, oval, kidney- or horseshoe-shaped nuclei and abundant cytoplasm (Fig. 2c). These tumor cells were positive for CD30 and their nuclei and cytoplasm were positive for ALK1 immunostaining (Fig. 2d). Considering these pathological results, we diagnosed the patient with recurrent systemic ALK-positive ALCL in the left fifth rib, at the primary site of the left maxillary bone. We decided not to surgically remove the lesion on the left fifth rib because the surgery would have been too invasive. Consequently, we followed the ALCL99 standard risk protocol (MTX3 arm) to treat the patient [8]. Although we found that the maxillary bone lesion was also an ALCL, we did not change the treatment because of the absence of high-risk organ involvement. MRI scans were performed several times to evaluate the lesions during treatment. The tumor size decreased from 30 × 30 mm to 16 × 20 mm, and a partial response was observed. A positron emission tomography-CT scan showed no significant accumulation, and a head CT scan revealed no maxillary bone lesions during hospitalization. At the time of writing, the patient has been progression free for 31 months since the completion of treatment.

Discussion

In this study, we encountered a case of primary bone ALK-positive ALCL that relapsed with rib metastasis after spontaneous remission without chemotherapy, following a biopsy of a maxillary bone lesion. Spontaneous remission of ALCL has been frequently reported in primary cutaneous ALCL [3] but is rare in systemic ALCL. In Table 1, we summarize 5 reported cases of spontaneous remission of systemic ALCL after biopsy, in addition to our case. The first 2 cases (nos. 1 and 2) mainly had superficial lymph node lesions [3, 4]. The remaining 3 cases (nos. 3, 4, and 5) involved deep lymph nodes, organs, soft tissue, etc. [5–7].

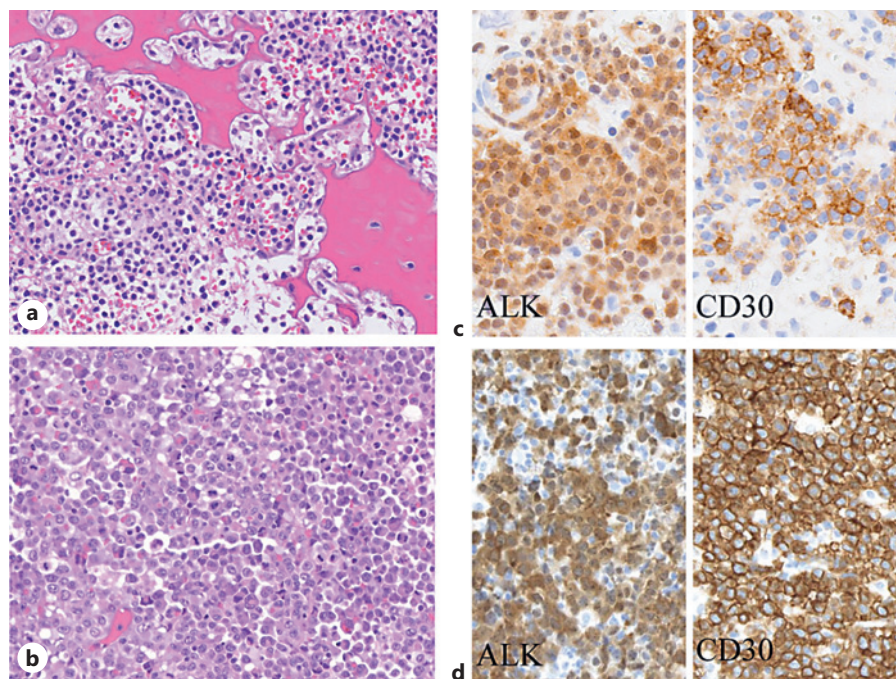


Fig. 2. Histology at diagnosis of primary and recurrent diseases. **a** Hematoxylin and eosin-stained section of the lesion on the left fifth rib. Atypical large cells with densely stained and swollen nuclei and abundant eosinophilic cytoplasm with occasional mitotic figures were observed. **b** Immunostained section of the lesion on the left fifth rib. The tumor cells were immunoreactive for CD30, and the nucleus and cytoplasm were positive for ALK1. **c** Hematoxylin and eosin-stained section of the lesion of the maxillary bone. A sheet-like proliferation pattern of large cells with a round, oval, kidney- or horseshoe-shaped nucleus and abundant cytoplasm was observed. **d** Immunostained section of the lesion of the maxillary bone. The tumor cells were positive for CD30 and ALK1 (nuclear and cytoplasmic).

These 5 cases were similar to ours in that the lesions spontaneously remitted after biopsy. In particular, in the 3rd, 4th, and 5th cases, the lesion remained after biopsy but the residual tumor spontaneously remitted, which was similar to our case. All 3 patients evaluated for ALK expression were ALK positive, as was our case. The mechanism of spontaneous remission of lymphoma remains unclear but may be due to changes in the host immune system associated with concurrent infection or biopsy. A traumatic intervention may activate the local immune system through cytotoxic T lymphocytes and therefore remit the lesion [9]. Thus, we cannot underestimate the relationship between biopsy and spontaneous remission. The mechanism behind spontaneous remission of systemic ALCL should therefore be further elucidated.

Primary bone lymphoma (PBL) is a very rare subtype that can be defined as a single bone lesion with or without local lymph node involvement. Additionally, it can present as several bone lesions lacking lymph node or visceral involvement [10]. The incidence of PBL is estimated to be approximately 5% of extranodal lymphomas and less than 1% of all NHL cases in adults and children [11]. The most common histological type of PBL is diffuse large B-cell lymphoma, whereas primary bone ALCL, the histological type of systemic ALCL as PBL, is extremely rare [10, 12, 13]. The pathogenesis of primary bone ALCL, especially the clinical significance of ALK expression, remains to be elucidated. To our knowledge, no case of spontaneous remission of PBL has been reported in the literature to date; however, our case is valuable in showing that bone lesions in primary bone ALK-positive ALCL can remit spontaneously.

Table 1. Cases with spontaneous remission of systemic ALCL after biopsy

Patient number	Age, years	ALK expression	Primary involvement	Biopsy method	Biopsy site	Time to relapse	Relapse involvement	Chemotherapy after relapse	Outcome	References
1	38	(+)	Left axillary lymph nodes	Excisional biopsy	Left axillary lymph node	(-)	(-)	Not applicable	Alive	[3]
2	12	(+)	Right axillary lymph nodes	Not mentioned	Right axillary lymph node	8 years	Left inguinal lymph node	(+)	Alive	[4]
3	44	Not mentioned	Left inguinal, paraaortic, hilar, and mediastinal lymph nodes Both lungs Stomach Left supraclavicular region	Not mentioned	Left inguinal lymph node Lung Stomach	2 months	Lymph node	(+)	Alive	[5]
4	77	Not mentioned	Left-sided soft palate	Incisional biopsy	Left-sided soft palate	12 months	Lymph node	(+)	Alive	[6]
5	35	(+)	Left axillary lymph node Subclavian vessels Thoracic and lumbar spine pelvis	Incisional biopsy	Left axillary lymph node	(-)	(-)	Not applicable	Alive	[7]
Our case	15	(+)	Maxillary bone (PBL)	Scraping	Maxillary bone	12 months	Left fifth rib bone	(+)	Alive	
PBL, primary bone lymphoma; ALK, anaplastic lymphoma kinase.										

Lymphoma, including ALCL, presents with nonspecific symptoms, as lymphadenopathy and fever are common inflammatory diseases that share many of the same features. Particularly in bone lesions, if symptoms improve, a detailed pathological examination for the diagnosis of lymphoma may be omitted because PBL is extremely rare. The prognosis of primary bone ALCL is not yet known; however, ALCL should always be considered since systemic ALCL is generally aggressive and there is a risk of relapse even after spontaneous remission, as in our case. Furthermore, a precise pathological diagnosis is vital in the differential diagnosis of primary bone lesions.

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Statement of Ethics

Written informed consent was obtained from the patient and his guardians for publication of this case report and any accompanying images. Ethical approval is not required for this study in accordance with local or national guidelines.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

K.A., F.Y., H. Shima, T.K., and T.I. treated this patient. H. Shimada managed and supervised the clinical practice. K.A. wrote the manuscript. F.Y., H. Shima, A.N., and H. Shimada reviewed and edited the manuscript. A.N. provided the figures. All authors contributed to discussions and agreed on the final version of the submitted manuscript.

Data Availability Statement

All data generated or analyzed during this study are included in this article and its online supplementary material. Further inquiries can be directed to the corresponding author.

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